

ORIGINAL ARTICLE

Primary care nursing activities with patients affected by physical chronic disease and common mental disorders: a qualitative descriptive study

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Aims and objectives. To describe nursing activities in primary care with patients affected by physical chronic disease and common mental disorders.

Background. Patients in primary care who are affected by physical chronic disease and common mental disorders such as anxiety and depression require care and follow-up based on their physical and mental health condition. Primary care nurses are increasingly expected to contribute to the care and follow-up of this growing clientele. However, little is known about the actual activities carried out by primary care nurses in providing this service in the Province of Quebec (Canada).

Methods. A qualitative descriptive study was conducted. Data were obtained through semistructured individual interviews with 13 nurses practising among patients with physical chronic disease in seven Family Medicine Groups in Quebec (Canada).

Results. Participants described five activity domains: assessment of physical and mental health condition, care planning, interprofessional collaboration, therapeutic relationship and health promotion.

Conclusion. The full potential of primary care nurses is not always exploited, and some activities could be improved. Evidence for including nurses in collaborative care for patients affected by physical chronic disease and common mental disorders has been shown but is not fully implemented in Family Medicine Groups. Future research should emphasise collaboration among mental health professionals, primary care nurses and family physicians in the care of patients with physical chronic disease and common mental disorders.

Relevance to clinical practice. Primary care nurses would benefit from gaining more knowledge about common mental disorders and from identifying the resources they need to contribute to managing them in an interdisciplinary team.

Key words: activities, common mental disorders, physical chronic disease, practice nurse, primary care

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What does this paper contribute to the wider global clinical community?

- This paper shows that even though nurses currently perform many activities with this clientele, they need to be more involved in the detection and followup of CMDs in primary care.
- Interprofessional collaboration including primary care nurses was found to be sub-optimal for the complexity of care required for people with physical CD and CMDs.

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Introduction

Common mental disorders (CMDs) such as anxiety and depression are two to three times higher among people with physical chronic disease (CD) in the general population (Scott *et al.* 2007). This relationship between CMDs and physical CD is linked to higher use of healthcare services, severe impairment, and decreased quality of life (Sareen *et al.* 2005, Kroenke *et al.* 2007, Teesson *et al.* 2011). The co-occurrence of these disorders requires that primary healthcare professionals address both physical and mental health conditions (Barnett *et al.* 2012).

Background

Patients in primary care who are affected by physical CD and CMDs such as anxiety and depression require care and follow-up based on their physical and mental health condition. Primary care nurses are increasingly expected to contribute to the care and follow-up of this growing clientele (Waterworth *et al.* 2015). Moreover, due to their skills in assessing physical, mental and psychosocial conditions, primary care nurses are well positioned to influence and bring innovative change to the care of people with physical CD and CMDs (Currid *et al.* 2012, Waterworth *et al.* 2015). With appropriate resources and training, primary care nurses could provide screening for CMDs using standardised tools, psychosocial support and health promotion activities (e.g. stress management training, support in changing lifestyle habits) (McKinlay *et al.* 2011). They could also provide telephone support for clients presenting depressive and anxiety symptoms (McKinlay *et al.* 2011).

Furthermore, a valid collaboration model for the treatment of depression for people with physical CD has shown that primary care nurses can be included in the care for this clientele (Katon *et al.* 2010). In fact, when primary care nurses have adequate support from a mental health specialist (psychologist or psychiatrist), work in collaboration with the family physician and deliver self-management support for physical CD and CMDs, both disorders can be improved (Katon *et al.* 2010). Collaborative care for the treatment of depression and anxiety has proven to be effective in primary care (Archer *et al.* 2012). Improvement in symptoms of depression and anxiety, better quality of life and higher patient satisfaction have all been shown (Archer *et al.* 2012, Coventry *et al.* 2014, Watzke *et al.* 2014). This type of care involves the presence of a family physician, a mental health specialist (e.g. psychiatrist) and a care manager (Archer *et al.* 2012). Having a primary care nurse as care manager for the treatment of depression for patients with physical CD

can improve depressive symptoms and increase patient satisfaction (Ekers *et al.* 2013). The care manager ensures regular contact with the patient, coordinates care with the family physician or mental health specialist, monitors the evolution of the patient's condition and adherence to treatment, initiates change when treatment is not effective and offers psychosocial support (Katon *et al.* 2001, Von Korff & Goldberg 2001).

To date, little is known about the actual activities carried out by primary care nurses in providing such care in the Province of Quebec (Canada). Without such knowledge, it is difficult to optimise and develop the practices of nurses. Additionally, despite what is known about the effectiveness of collaborative practice and the use of nurses as care managers for this clientele, few studies have shown whether primary care nurses actually perform these activities in day-to-day practice. The purpose of this study was to describe the activities of primary care nurses practising with patients affected by physical CD and CMDs in primary care settings in the Province of Quebec.

Methods

Design

A qualitative descriptive study was conducted (Sandelowski 2000). The study builds on a research project by Roberge *et al.* (2016) aimed at exploring 'the perceived needs, barriers and facilitators for the delivery of mental health care for patients with coexisting CMDs and physical CD in primary care from the clinician and patient perspectives', through interviews with primary care nurses, family physicians and other healthcare providers. To further explore nursing practice with this population, we expanded on the Roberge *et al.* (2016) study by recruiting a larger sample of nurses from a range of clinics in primary care.

Setting

The study was conducted in Family Medicine Groups (FMGs). Launched as an official designation in the early 2000s, there are now about 250 FMGs in the Province of Quebec, Canada (Leclerc 2015). An FMG is a group of family physicians who work in collaboration with primary care nurses and other health professionals to offer family medicine services. Part of their mission is to improve access to care and services, notably by promoting the use of primary care nurses and family physicians to support people with CD (Émond *et al.* 2003). Primary care nurses in FMGs are increasingly involved in such care.

Sample and recruiting

A purposeful sample was recruited (Patton 2015). We set the minimum sample size at 12 nurses for a reasonable balance between feasibility and an acceptable degree of data saturation (Grove *et al.* 2013). There were two criteria: (1) that the nurses be involved in caring for patients with CD and (2) that they have at least one year of clinical experience. These criteria ensured that the nurses in our sample had the appropriate experience and the ability to provide concrete examples for our study.

Recruiting was done in two phases. In the first phase, Roberge *et al.* (2016) started with three FMGs, each being in a different region of Quebec. At each FMG, a meeting was held with the stakeholders concerned and an information pamphlet was distributed to care providers and patients. This first phase led to six nurses agreeing to participate in the study. Due to the variability in nurses' activities from one FMG to another, the research team decided to continue recruiting, both to enlarge the sample of nurses and to ensure that they came from a wider range of FMGs.

The second phase of recruiting involved six additional FMGs. After obtaining the required approvals, nurses were contacted by telephone about the project, and those who expressed interest in participating were emailed further information. Seven more nurses were recruited in this phase, from four of the six FMGs. Ethics approval was obtained from the ethics committees of the Centre de santé et de services sociaux de Chicoutimi, the Centre hospitalier de St-Mary and the CSSS – Institut universitaire de gériatrie de Sherbrooke.

Data collection

Data were collected using semistructured individual interviews and a sociodemographic questionnaire that included occupational questions. The interview guide was based on Coventry *et al.* (2011) and on nursing activities suggested in the literature for this clientele. The guide addressed four topics: screening for CMDs, acts and activities, self-management support and interprofessional collaboration. Ranging from 45–60 minutes in length, the interviews were conducted between June 2014–August 2015 by an experienced psychologist ($n = 6$; phase 1) and by the lead researcher ($n = 7$; phase 2). Except for three done by telephone, the interviews took place in the office of the nurse concerned. The lead researcher also conducted validation interviews with interested participants ($n = 7$) to validate the preliminary results and to deepen our understanding of certain activities. The validation interviews took 15–30 minutes and were done either in person or by telephone, depending on the preferences of each participant. Audio recordings were made of all interviews, and the recordings were then

transcribed. As the interviews were done in French, all quotations presented in the Results section of this paper have been translated to English.

Data analysis

The interviews were analysed using a mixed coding method based on the principal themes in the interview guide and on the emerging data (Miles *et al.* 2014). The analysis process was guided by an evidence-based structure for describing the professional practice of nurses (Poitras *et al.* 2016). In this structure, three main concepts are used in describing nursing practice: professional role, domain and activity. Thus, activities were identified and then grouped into activity domains. The software used for data management was NVIVO version 10; QSR international. Three researchers (AG, MCC, MEP) analysed the first interview, and all subsequent interviews were analysed by AG and MCC. Four meetings were held from March–August 2015 in which all the researchers came together to discuss emerging themes, data organisation and content.

Results

Data analysis enabled us to list the activity domains in which FMG nurses provide care for patients with physical CD and CMDs. The following paragraphs provide a detailed description of the context and activities of each domain.

Context

Sample characteristics

A total of 13 nurses from seven FMGs agreed to take part in the study. All participants were female, most having completed a Bachelor of Science in Nursing ($n = 11$). Of those without a nursing degree, one held a certificate in management and one had completed some university courses in nursing sciences. Most of the participants were in their thirties ($n = 7$) (Table 1).

Mental health training

Most of the nurses ($n = 9$) had received up to two half-days of continuing education on mental health within the past year. Four nurses from one FMG had recently been involved in the development of a low-intensity tool for evaluating and following patients with CMDs. Thus, they had received training on screening and following patients with CMDs in primary care.

Access to psychosocial services

Seven nurses had access to a psychologist and/or a social worker in their clinic. At three FMGs (2-4-6), mental health

Table 1 Characteristics of participants

Nurse	Age group (years)	Years of experience as a nurse	Years of experience in Family Medicine Group
PCN_1	30–39	5–10	≤4
PCN_2	30–39	11–16	≤4
PCN_3	18–29	5–10	5–7
PCN_4	30–39	5–10	5–7
PCN_5	30–39	11–16	8–10
PCN_6	40–49	17–24	≤4
PCN_7	40–49	17–24	8–10
PCN_8	30–39	11–16	5–7
PCN_9	40–49	11–16	≤4
PCN_10	30–39	5–10	≤4
PCN_11	40–49	>25	≤4
PCN_12	30–39	5–10	≤4
PCN_13	50–59	11–16	≤4

professionals (social workers or psychologists) were available for patients approximately one day per week.

Not all FMGs have the benefit of such professionals, either within the clinic or off-site, and some participants reported that they could not refer patients to a psychologist, but only to a social worker, who would then refer the patient for psychological support if needed. Table 2 presents the characteristics of the FMGs that took part in the study.

General involvement of nurses in managing patients with CMDs

Most of the participants said they rarely followed patients solely for CMDs. Generally they followed patients with CMDs during follow-up on their physical CD, for example ‘One of the things I’ve learned: I’ll tell them to come for lifestyle training and diabetes, and end up following them for their mood, along with their diabetes’ (PCN_6). A few participants mentioned that a physician in their FMG might occasionally refer a patient to them for monitoring CMDs, with or without physical CD, but this was rare. They noted, however, that they often worked with patients who presented CMDs or psychological distress during walk-in clinics, and not necessarily in the presence of physical CD.

Domains of activity

The activity domains described in this section are presented in logical rather than chronological order. In providing care for physical CD and CMDs, nurses assess the patient’s physical and mental health condition, plan for care, collaborate with other professionals, sustain a therapeutic relationship and engage in health promotion activities (Table 3). The sequence of activities varies from one visit to another.

Assessment of physical and mental health condition

This domain includes the following activities: data collection (whether by interviewing the patient or using data collection tools) and monitoring mental health.

Data collection. Each time a nurse meets a patient for physical CD, her first task is to inquire on the patient’s general status. She remains attentive to the patient’s overall experience, for example ‘My first question, *How are you doing?* isn’t about diabetes. They’ll say, *Not so good, I’m pretty down these days.* Which gives me a general idea’ (PCN_8).

Most nurses will note the patient’s mental health history as well as current medication. This information let them target at-risk patients and watch for changes in mental health while evaluating and monitoring their physical CD, for example ‘I always look at medical history; I look at their medication list. Often they’re already taking Effexor or something else for their mood, because it isn’t easy living with a chronic condition’ (PCN_11).

When the nurse notices a change in the patient’s mental health, she takes more detailed data for an in-depth evaluation of the patient’s physical and mental health. She will try to get bio-psycho-social data by asking the patient directly. She will also pay particular attention to the evolution of CMDs in a patient who has already been diagnosed:

Mood, social aspects, is there social withdrawal, environmental stressors, good sleep, good appetite? Weight loss? Does he cry regularly? If I see that he is a bit on edge ... Does he have social support from loved ones? Daily burden? Does he have to do everything on his own? And if I notice that he has less capacity, and things may be getting difficult, well we’ll dig around that. (PCN_12)

Table 2 Characteristics of Family Medicine Groups (FMGs)

FMG	1	2	3	4	5	6	7
Nurses (<i>n</i>)	3	3	8	2	2	2	4
Family doctors (<i>n</i>)	27	13	45	10	9	9	9
Patients (<i>≈n</i>)	12,000	10,000	30,000	10,000	12,000	12,000	16,000
Social Worker (<i>n</i>)	1	0	0	0	0	0	1
Psychologist (<i>n</i>)	1	1	1	0	0	0	0

Table 3 Activity domains and description

Activity domains	Activities
Assessment of physical and mental health condition	<i>Data collection</i>
	Inquire about the patient's general condition
	Question the patient directly about mental health condition
	Use tools for mental health condition and common mental disorders (CMDs)
	<i>Monitor mental health condition</i>
	Mood
	Suicidal ideation and degree of dangerousness
Care plan and nursing follow-up	CMD treatment effects
	<i>Establish action priorities</i>
	Refer the person to emergency services
	Develop an interdisciplinary care plan
	Develop an intervention plan with the patient
	<i>Adapt follow-up to patient</i>
	Telephone reminder
Interprofessional collaboration	Telephone follow-up
	Allow for more time in scheduling visits
	Refer the patient to other care professionals
	Consult other care professionals
Therapeutic relationship	Share information with other care professionals
	Establish trust
	Reassure the patient
	Normalise experience
	Draw patient's attention to external resources
Health promotion	Active listening
	Accompaniment through acceptance of the disorder
	Encourage and highlight progress in health behaviours
	Reinforce autonomy
	Enhance patient responsibility for choices
	Basic training on pathologies and lifestyle habits
	Provide written information on CMDs
	Reinforce and validate knowledge on CMDs

The use of data collection tools in assessing mental health and screening for CMDs varies from place to place, possibly due to different levels of training and peer support. Four nurses had access to an efficient mental health assessment tool that helped them to detect anxiety and depressive symptoms. When symptoms are detected, they use the Patient Health Questionnaire (PHQ-9) for depression, or the Generalised Anxiety Disorder (GAD-7) for anxiety. Some nurses are in the process of developing assessment tools, while others use no assessment or screening tools at all.

Monitoring mental and physical health. When a patient has been diagnosed with a CMD by a physician, or presents psychological distress while being followed for a physical condition, the nurse will pay attention to the person's mood, watching for the presence of suicide ideation and the degree of dangerousness. Additionally, even if the physician has not requested it, most nurses will assess adherence to ongoing treatment for CMDs as well as any side effects:

I also look at side effects. Sometimes they'll start taking antidepressants and they're out of it, drowsy. That's normal at the start of treatment, but after two weeks, the effect should have abated. So I'll look at that. (PCN_8)

Planning for care and follow-up

Care planning is the activity domain in which the nurse ensures that the patient receives appropriate care for his or her situation. This requires setting priorities. Decisions will be made in consultation with the patient, taking into account their health priorities and the pace at which they wish to proceed: 'We'll talk with the patient about what he needs... If what's important to him is stabilizing his diabetes and then losing weight, we'll go with that'. (PCN_4)

Establish intervention priorities. Our participants reported establishing short- and long-term priorities with patients. For example, if a patient presents suicide ideation, the nurse ensures that that he or she is quickly taken in charge by a psychosocial emergency service or a physician. Over the longer term, when care becomes more complex, nurses may follow an individualised care plan with an interdisciplinary team, or one agreed upon between nurse and patient, for example 'Sure, ideally... when there are major issues, individual and interdisciplinary care plans, we call in every caregiver the patient sees, and try to have a common plan' (PCN_15). Intervention priorities depend on the patient and complexity of the situation, as well as the resources available to the nurse.

Adapting follow-up to the patient. Nurses stressed the importance of adapting follow-up visits to the needs and profile of the patient. This is crucial for patients with physical CD and CMDs or psychological distress, because as our participants reported, a visit can go in a completely different direction than planned:

The man with diabetes, that other time... I put the device away, I just focused on the emotional aspect. Diet and physical activity, that would be for the next visit. Because it was more than he could handle, so I thought, even if I throw myself into showing him how to eat well, he's still stuck in a state of shock... (PCN_11)

The nurses also mentioned that they allotted more time in their schedule for patients presenting physical CD and CMDs or psychological distress:

Like for diabetes, those will be one-hour appointments, but if I see that next time I'll be seeing him for blood pressure, or if there was an issue, I'll reserve more time, because I know the person will probably talk a bit more, so the conversation might veer off. (PCN_9)

Additionally, when patients presented a more precarious mental health condition, some of the nurses would offer to call them at home more often (regarding their physical CD) so they wouldn't have to travel as much. Some patients prefer not to get help for their CMDs. In that situation, the nurse may call the patient by telephone: 'For sure we have to follow up anyway, at least call them on the telephone, those clients, to see how they're doing' (PCN_5).

Interprofessional collaboration

Three forms of collaboration were often used by our participants: referral to another professional, consulting an expert and sharing information about the patient's condition with another professional. Forms of collaboration can vary from one FMG to another and tend to be optimal when all members of the interdisciplinary team work in the same clinic. Our participants noted the benefits of face-to-face contact with other health professionals who were following their patients.

Referral to other care providers. It seems to be common for nurses to refer patients with CMDs to other healthcare professionals, not always because the care required is beyond the nurse's legal scope of practice. Sometimes the referral is due to lack of time, or because the nurse feels that she lacks the knowledge to provide adequate care for CMDs:

As for time, well, I mean in a crisis situation, or a problem, yes I feel I have the tools, we do a lot of listening... But for the rest, to manage anxiety, management and all that, we refer our patients. (PCN_7)

Some will quickly refer a patient to another care provider, often a social worker, and not go any further in treating the CMD. That is, depending on the extent of the problem and the time required for another professional to take charge, she may not deal with the CMD at all while working with the patient for his or her physical CD:

I don't really follow patients for CMDs as such, but when we detect things, of course we'll refer a patient to the hospital's front-line social services. You know, I refer them and I rely a bit on the social worker when he meets them. But if not... it's like my focus is more on the diabetes. (PCN_10)

At all the FMGs, participants reported that they often referred patients to their family physician. When possible, if they felt that medical assessment was needed and that treatment should begin quickly, they would telephone or email the physician. They would even do this if a patient came to their office and was in crisis and needed immediate medical care. The treating physician was their first resource in the event of complications or a deterioration in the patient's mental health.

Consult another professional. Participants also reported that they often consulted other professionals for specific information related to a patient's CMD, or for general information about CMDs and how to treat them. Usually they would first consult the patient's physician, then other professionals if necessary. For example, some mentioned that they would consult the local pharmacist if they had questions about drug interactions or side effects reported by the patient. When they had doubts about a patient's adherence to treatment, some would call the pharmacist to get his impression:

The pharmacist sees the client as often as I do, in the sense that the client goes and gets his medication there, so the pharmacist knows him quite well. And clients often consult pharmacists, so they know them in a certain way. When it's a complex case, often I'll call the pharmacist to discuss medication, adherence to treatment: Is he still renewing his prescriptions? To see what [the pharmacist] thinks. (PCN_5)

Sharing information. Information sharing is easier in FMGs due to the proximity of physicians and nurses. They inform each other of any changes in the physical and mental health of patients they are following. For example, when a nurse sees a patient for physical CD and notices a change in mood or behaviour, she will inform the physician by email or through progress notes:

The patient clearly tells me when I ask him, it's not going well, he's depressed, his symptoms are worsening... Or sometimes without making an appointment for the patient I'll speak with their physician to see what he thinks – Oh yes, have him see me, you know, or sometimes, I ask questions and tell [the physician] about my interview with the patient, and I'll ask for his opinion. (PCN_8)

In one FMG, where a tool had been developed for patients with mental health conditions, the nurses would complete their evaluation and then discuss the appropriate care, treatment and follow-up with the physician.

In addition to the physician, one participant mentioned that occasionally the psychologist would call to tell her what she was working on with the patient, in terms of the

latter's current mental health status: 'So sometimes, I talk to [the psychologist] and we exchange views about the patient, then she says OK, so *this is what I'm working on with him, can you provide additional support?*' (PCN_2). This was not very common, however. Many of the nurses mentioned that when they received a report from a social worker or psychologist, it was sent to them by fax.

Lastly, some of the nurses reported sharing information with mental health nurses from the local community services centre who were also following patients with CMDs.

Therapeutic relationship

Throughout the treatment of physical CD, over many appointments, patient and nurse develop a trust-based relationship through which the nurse comes to know the patient and can recognise significant changes in mental health. This relationship facilitates the exchange of information and predisposes patients to express themselves more openly:

I enter their private world, not because I want to, I become their confidant if you will, because we're going to deal with subjects... So, how's it going with your diabetes, your medication, the side effects, rest, stress? ... And spontaneously, sometimes they'll talk to me: Oh, my husband passed away. (PCN_11)

In large part, this relationship of trust develops by putting oneself at the same level as the patient, sharing common objectives and displaying empathy: 'I think that as soon as they have our attention and we really listen to what their needs are and what they want, well it's super easy to work with them afterwards and establish a trusting relationship' (PCN_4). Once that trust is established, a therapeutic relationship is formed through which nurses can more easily support patients with their trials and challenges. For example, our nurses reported reassuring their patients, normalising experienced events and drawing their attention to the resources available.

Most of our participants stressed that they did a lot of active listening with patients who had physical CD or CMDs. Some pointed out that some patients are socially isolated and feel stigmatised, thus the importance of careful listening.

Some of the nurses acknowledged that for some patients, being diagnosed with a physical CD can come as a shock, and be a precursor to anxiety-related or depressive symptoms. One participant mentioned being more vigilant for signs of psychological distress in such cases, and ensuring a shorter interval between follow-up visits:

Recently, I had a lady who was distraught because she'd recently been informed that she was diabetic... I offered her a second and third meeting, sooner than usual. I make sure that each time she comes, our session goes well, there aren't as many tears, and I ask

her how things are going, if she's still crying, if she's started talking about it with her family. (PCN_12)

In this manner, nurses accompany the patient in accepting his or her illness, exploring emotions and apprehensions towards the new disease. Some nurses mentioned having to do the same with respect to accepting a CMD diagnosed in the course of care for a physical CD.

Health promotion

Nurses support patients by providing encouragement and highlighting their progress in health behaviours from visit to visit. They try to empower patients in their choices, reinforcing their autonomy in managing both the physical and mental aspects of their CD. One nurse said that when possible, she includes a family member to help the patient in this process. Some of our participants had received training in motivational interviewing as a strategy to use with patients with CMDs and physical CD.

Educating patients about CMDs bears no resemblance to formal instruction, being an activity that is carried out spontaneously and repeated over the course of care for physical CD, in line with the patient's immediate needs. The nurses mentioned that inevitably they come to educate the patient about his or her CMD, and if desired, validate and reinforce the patient's understanding. They noted that many people are reluctant to talk about CMDs, and false beliefs are still very common:

Often, it's still not that well known, it's taboo, mental [illness], and when it's new, the first episode, well they'll ask lots of questions, Why me? They have difficulty understanding, so I explain what goes on in the brain, that helps too. (PCN_5)

Some of the nurses also provided printed matter about CMDs and discussed them openly with the patient. As with physical CD, our participants emphasised that they teach healthy lifestyle habits (proper sleep hygiene, physical exercise, reducing tobacco, alcohol, drugs, etc.) and explain ways to manage stress:

I think where I can help someone the most with a mental health problem is with lifestyle habits: proper sleep hygiene, start doing physical exercise, even if it's just 15 minutes, change their mindset, try to have a good schedule. I'll emphasize those things. (PCN_2)

However, our participants reported being better equipped to conduct teaching activities related to physical CD than they are to CMDs. Preventing CMDs in patients with physical CD does not seem to be a common health promotion activity. In fact, one nurse asserted that she did not really discuss psychological complications related to physical CD:

Patients are not aware of this, that there's a correlation between the two. Often, we'll address all the physical complications that can be brought on, diabetes, hypertension, whatever, but we rarely address the mental health issues that come with it. (PCN_1)

Discussion

The objective of our study was to describe the activities of FMG nurses working with patients who have physical CD and CMDs. Well-developed and patient-focused, nursing practice in CD consists of many nursing activities and skills (Chouinard *et al.* 2015), which are specified in the legislative framework of Quebec and Canada (Canadian Nurse Association 2015; Ordre des infirmiers et infirmières du Québec 2016). Our results show that when primary care nurses work with patients who have physical CD, they also perform a variety of activities in connection with CMDs. These activities are variable and appear to be influenced by various factors that have yet to be specified.

Activities undertaken for the CMDs of patients with physical CD vary from nurse to nurse and between different FMGs. Such activities may deploy clinical tools and clear directives in a particular FMG, or simply be based on the nurse's knowledge and clinical judgement, with no clear direction from superiors. Waterworth *et al.* (2015) also noted variability in the means by which nurses attempt to detect depression and to provide support when depression accompanies physical CD. There are many factors behind this variability, including practice context and the different approaches in different settings towards integrating physical and mental health care by an interdisciplinary team (Maxwell *et al.* 2013). Variability in the knowledge and confidence of nurses who work with patients presenting CMDs can also explain some of the variability in the activities they perform (Waterworth *et al.* 2015).

The first step in the treatment and management of depression in a person with physical CD is the detection or recognition of anxiety and depressive symptoms (National Institute For Health and Clinical Excellence, 2009). In our study, most participants were vigilant for signs of psychological distress in patients with physical CD, as well as for declining mental health. Four nurses were also aware of the importance of detecting anxiety and depressive symptoms in people with physical CD, and of screening for CMDs when symptoms are detected.

Screening for CMDs when anxiety and depressive symptoms are detected, with or without a standardised tool, is more effective when there is a team whose mandate includes the care of patients with CMDs (Joffres *et al.* 2013). Screening for CMDs can be awkward for nurses who lack the skills to assist the patient after screening (Maxwell *et al.* 2013). In contrast, nurses who have been trained in mental health,

who have a standardised tool that is appropriate for their place of work, and who are supported by a psychologist and physicians, are more at ease with screening and providing care for CMDs. Our study corroborates the observations of McKinlay *et al.* (2011) and Waterworth *et al.* (2015), who share the opinion that primary care nurses are well positioned to screen for CMDs, but in the absence of collaborative practices and specific skills it is difficult for them to perform this task as fully as they might.

Another gap between potential and reality has to do with providing support for patient self-management of CMDs, which did not come up spontaneously in the interviews with our participants. Yet this activity is known to be effective when delivered by a primary care nurse who has the necessary resources and training (Scanlan *et al.* 2006, DeJesus *et al.* 2014). What did come up had more to do with support for self-management of physical CD. It is quite possible that, as with detection in primary care, support for self-management of CMDs could be facilitated by training and the development of specific tools.

Despite evidence of the benefits of a collaborative approach in treating CMDs in primary care (Katon *et al.* 2010, Archer *et al.* 2012, Coventry *et al.* 2014), it remains infrequent and nurses are rarely involved. Yet when we compare the activities of a collaborative care manager (Von Korff & Goldberg 2001) with those of an FMG nurse, we note that most of what the former does consists of activities that are often performed by nurses: monitoring the patient's health status and adherence to treatment, taking action when treatment is not effective, ensuring regular contact with the patient and providing psychosocial support. However, the coordination of care and services in mental health, an important aspect of the role of a care manager, did not emerge as an activity conducted by the nurses in our study. Besides the need for changes in collaborative care practice, for nurses to fully assume the role of care manager for patients with CMDs and physical CD they must have more training in providing mental health care in the FMG setting.

Our study highlights the fact that notwithstanding the complex care involved in treating patients with physical CD, the types of collaboration described by nurses are mainly in parallel and by referral/consultation as defined by Careau *et al.* (2014). Yet for any complex clientele, a shared practice should be the norm, with all involved having a common goal with regard to patient needs and the needs of the family (Careau *et al.* 2014). Although such collaboration varies from one FMG to another, it seems to be optimal when the psychologist and social worker are also in the clinic, and when nurses are appropriately trained and have standardised tools for CMD screening. Collaboration between nurses and physicians was also found to be an essential factor.

Strengths and limitations

Our description of the activities of FMG nurses with patients who have physical CD and CMDs provides a first picture of what is actually done by such nurses in FMGs in Quebec. The number of FMGs included ($n = 7$) enabled us to document a wide range of nursing activities. Best practices were identified in one FMG, offering a good example for other FMGs. However, the data were collected through qualitative interviews with nurses only. The latter may have forgotten certain elements, or may have given an enhanced portrait of their practice. Observation of nurses interacting with patients with physical CD and CMDs would have strengthened the credibility of results. Interviews with patients and their families would have allowed us to explore their perception of the care they receive from FMG nurses. Lastly, a larger sample might have revealed other activities, such as the nurse's role in involving the family in the care of patients with physical CD and CMDs.

Conclusion

Our results suggest that there are disparities among different FMGs as to the nursing activities performed. Certain activities that are essential to providing care for this clientele are under-developed in Quebec FMGs. For example, CMD screening and providing support for self-management were not generally done by the nurses in our study. Also, interprofessional collaboration and care coordination by FMG nurses were not as good as they could be to meet the complex needs of patients with physical CD and CMDs. Although nurses currently perform many activities with such patients, they are not sufficiently involved in the detection and management of CMDs in primary care. Based on the evidence, there is a gap between the kinds of activities expected of primary care nurses and what

they actually do when CMDs are present in patients with physical CD.

Thus, improvement is needed in clinical practices, and FMG stakeholders should strongly consider the use of primary care nurses in collaborative practice models for CMDs.

Relevance to clinical practice

Primary care nurses would benefit from receiving more training about CMDs and from identifying the resources needed to follow patients affected by physical CD and CMDs, whether alone or as part of an interdisciplinary team. Many primary care nurses are already involved in following this clientele and would benefit from the expertise of other healthcare professionals to optimise their delivery of crucial activities. The full range of activities they perform should be recognised and shared with other nurses practising in primary care settings.

Contributions

All authors contributed to the study design, the data collection and analysis, and to the manuscript preparation.

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Conflict of interests

The authors declare no conflict of interests.

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